Heterogeneous UAVs Trajectory Optimization for Post-Disaster Target Search Based on MARL With Graph Attention Network

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***Abstract*—Large-scale post-disaster target search missions pose significant challenges for Uncrewed Aerial Vehicles (UAVs) due to the complex and unpredictable environments. Heterogeneous swarms that combine the benefits of multiple UAVs can enhance adaptability and search efficiency. Therefore, a leader-follower strategy is employed for heterogeneous multi-UAVs in this paper, where fixed-wing UAVs serve as leaders to provide dynamic relay communication coverage, while multi-rotor UAVs act as followers for multi-target search tasks. We construct a cooperative graph to model relationships among neighboring UAVs and design importance indices for both cooperation and search targets to optimize this graph, streamlining state observation vectors. We then introduce Graph Attention Networks (GAT) to aggregate information from adjacent nodes in the cooperative graph. Additionally, we employ Multi-Agent Reinforcement Learning (MARL) to optimize flight trajectories separately for fixed-wing and multi-rotor UAVs, enhancing cooperation efficiency in multi-target searches. Based on the above methods, we propose a Graph Attention Network multi-agent Actor-Critic (GATAC) algorithm, which effectively addresses the dimension explosion problem that arises as the number of agents increases. Simulation results demonstrate that the GATAC algorithm outperforms existing methods in terms of target search, energy consumption and broken link duration. Finally, we establish a multi-UAV cooperative validation platform, and experimental trajectory plots confirm the stable performance of the GATAC algorithm in real-world environments.**

***Index Terms*—Graph attention, multi-agent reinforcement learning, multi-UAV trajectory optimization, multi-target search.**

I. Introduction

POST-DISASTER Search and Rescue (SAR) is a critical part of saving lives [1]. However, the post-disaster SAR process is usually characterized by unfavorable conditions [2], such as collapsed houses and extensive damage to ground communication equipment [3], [4], [5]. In such environment, Uncrewed Aerial Vehicles (UAVs) have been widely employed in post-disaster SAR missions in recent years due to their advantages of easy deployment, rapid response, and wide coverage area [6], [7]. However, due to the inherent constraints between UAV size and its endurance and payload capacity [8], larger-size UAVs struggle in confined spaces while smaller counterparts suffer from joint limitations in batter endurance and communication reliability [9]. It is often difficult to meet the post-disaster SAR mission requirements by employing a single type of UAVs. Heterogeneous UAV swarms combine the strengths of multiple types of UAVs [10]. This combination allows detailed searches in narrow areas while covering diverse mission needs, ensuring complementary performance and efficiency.

Furthermore, this paper presents a leader-follower cooperative model for heterogeneous multi-UAV systems, combining fixed-wing and multi-rotor UAVs. In this model, fixed-wing UAVs, with their large payload capacity and long endurance, can provide continuous dynamic relay communication services within mission-critical airspace. Additionally, they can quickly sense the approximate location of targets on the ground by carrying high-powered and wide-coverage detectors. The fixed-wing UAVs distribute this information to multi-rotor UAVs, thereby guiding multi-rotor UAVs for detailed searches [11], [12]. Multi-rotor UAVs exploit their agile maneuverability to perform post-disaster multi-target search tasks in complex and obstacle-laden environments.

Although the leader-follower multi-UAV cooperation architecture improves the adaptability of UAV swarms to the environment, cooperative control of UAV swarms is also a complex and challenging task [13]. With the increase of the heterogeneous multi-UAV scalability, communication among UAVs

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will become more complex, and problems such as rational task allocation and scheduling will arise [14]. In summary, the challenges of the leader-follower multi-UAV cooperation model are given as follows: 1) How to achieve cooperative trajectory optimization for heterogeneous multi-UAVs; 2) How to enhance the scalability of heterogeneous multi-UAV cooperation.

In the research on cooperative trajectory optimization for heterogeneous multi-UAV systems, model-based methods have been widely used [15], [16]. However, these methods require high modeling accuracy and suffer high computational complexity in their implementation [17]. Currently, Multi-Agent Reinforcement Learning (MARL) methods have been widely employed to solve the cooperative trajectory optimization of heterogeneous UAVs. The MARL methods mainly fall into value-based and policy-based approaches [18], [19], both requiring appropriate reward functions to guide agents in maximizing cumulative rewards through training [20]. To optimize reward maximization and cooperative policies through training, we use the Centralized Training and Decentralized Execution (CTDE) architecture, which is suitable for multi-agent systems [21], [22]. This approach utilizes global and communication information during training but relies exclusively on the local observations of the individual agent during execution [19], thereby enhancing the efficiency and cooperative capabilities of the overall system.

As the scale of heterogeneous multi-UAV cooperation increases, the environment becomes unstable, posing challenge to value-based methods. In contrast, the policy-based method directly learns the optimal policy, which is highly effective in unstable environments [23]. However, with the increasing number of agents, the CTDE-based algorithms can lead to dimensionality disasters, hindering effective model training [24]. To address these challenges, the graph structure is introduced in this paper, agents can be viewed as nodes and the connectivity between agents can be represented using edges. This structure effectively mitigates the redundancy of input information, thus increasing the efficiency and computational performance of the model. In summary, this paper makes the following main contributions.

1) To address the distant multi-target search problem in unknown complex environments, we present a leader-follower heterogeneous multi-UAV cooperative model. Then we propose the Graph Attention Network multi-agent Actor-Critic (GATAC) algorithm to optimize the heterogeneous UAV trajectories in the model to enhance UAV cooperation in large-scale missions.
2) To tackle the issue of dimension explosion in centralized training with increasing UAVs scale and target points, a graph attention model is integrated into the GATAC algorithm. By constructing a cooperation graph, the dimensionality of input vectors for each UAVs in both the critic and action networks is reduced, effectively enhancing the scalability of multi-UAV systems.
3) We conducted a large number of simulation experiments. The experimental results demonstrate that the GATAC algorithm has superior performance. Then, based on the motion capture system, a multi-UAV verification platform was built for physical verification to prove that the proposed GATAC algorithm can be effectively applied to practical scenarios. It realizes the long-distance and large-scale cooperative deployment of heterogeneous multi-UAVs in an environment without base station signal coverage.

II. Related Work

This section will review the related work from the following two aspects: 1) Cooperative target search by multi-UAVs; 2) MARL methods for UAVs trajectory planning.

A. Cooperative Target Search by Multi-UAVs

To address the cooperative search problem among multiple UAVs in tasks such as search, attack, and rescue, Zhang et al. [25] proposed a distributed cooperative search method based on Ant Colony Optimization (ACO) algorithm that enhances information consistency and cooperative efficiency. In rugged terrain scenarios, communication among UAVs and the ground devices may be obstructed by mountains and buildings. To tackle this issue, Savkin et al. [26] presented a joint multi-UAV path planning and transmission scheduling algorithm to address the communication scheduling problem among multi-UAVs and ground-deployed Reconfigurable Intelligent Surfaces (RISs). To address the fairness issue of multi-UAV-assisted Mobile Edge Computing (MEC) systems, He et al. [27] proposed a multi-UAV trajectory optimization method, achieving minimum energy consumption while ensuring task fairness and efficiency. In the above works [25], [26], [27], the multi-UAV systems primarily consist of homogeneous UAVs. However, constrained by factors such as UAV endurance, payload capacity, and communication range, the application scenarios of homogeneous multi-UAV systems are limited.

Heterogeneous multi-UAV systems have significant advantages in executing complex tasks and responding to diverse demands. For instance, fixed-wing UAVs are suitable for long-distance cruising and high-altitude surveillance, while multi-rotor UAVs are suitable for low-altitude hovering and detailed search tasks [28]. Such heterogeneous swarms are able to perform diverse tasks. Yu et al. [29] devised the Improved Genetic Algorithm (IGA) with a special unlocking mechanism to solve the task allocation problem, facilitating the deployment of reconnaissance UAVs, combat UAVs, and munitions UAVs in combination. Addressing the heterogeneous UAV coverage path planning problem, Chen et al. [30] proposed a UAV path planning algorithm based on the ant colony system, achieving the optimal path for UAVs to fully access and search all areas of interest. For heterogeneous platforms composed of UAVs and Uncrewed Surface Vehicles (USVs), Leng et al. [31] proposed a task coordination allocation method based on improved Discrete Particle Swarm Optimization (DPSO), capable of rapidly converging to the global optimal solution of the objective function. To tackle the trajectory planning problem of heterogeneous UAVs with different flight and scanning capabilities,

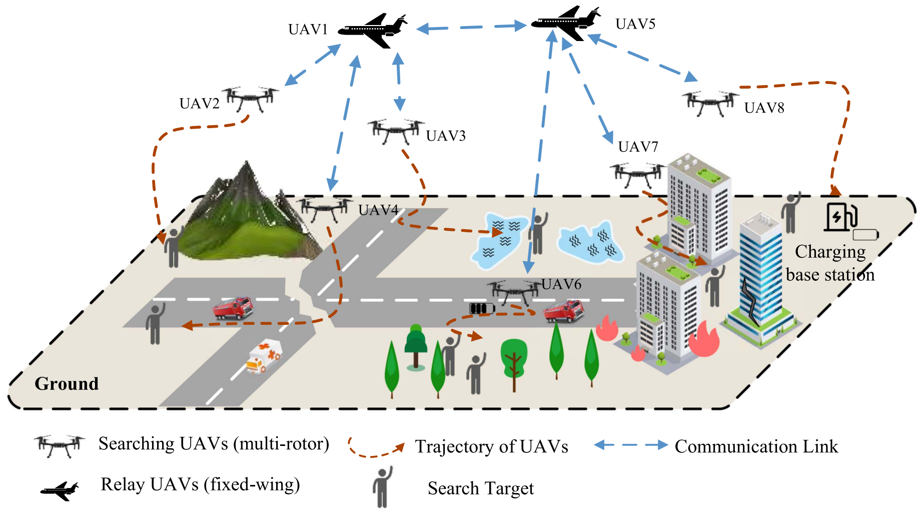


Fig. 1. Schematic diagram of emergency search and rescue scenarios with multi-UAVs.

Chen et al. [32] proposed a UAV trajectory planning algorithm based on symbiotic interaction behavior in biology. The above studies [29], [30], [31], [32] adopt bio-inspired or optimization-based approaches to address the coordination problems of heterogeneous UAVs, However, these methods are complex, with poor real-time/adaptability.

B. MARL Methods for UAVs Trajectory Planning

The MARL enables multi-UAVs to achieve distributed cooperative decision-making and control through autonomous learning and interaction without centralized controller, thereby enhancing system flexibility and robustness [33]. Currently, MARL algorithms are mainly classified into two categories.

The value-based method mainly learns the value function. Gong et al. [34] proposed a cooperative air combat strategy framework for multi-UAV systems, which is based on value decomposition and reward functions derived from expert cooperative air combat experience. For UAV Internet of Things (IoT) data collection, Li et al. [35] proposed a UAV trajectory planning algorithm based on DQN, aiming to minimize the average Age of Information (AoI) of SNs while satisfying UAV energy and collision avoidance constraints. To investigate the data packet routing problem in multi-hop UAV relay networks, Ding et al. [36] proposed a QMIX-based algorithm, which optimizes UAVs trajectories to improve routing in multi-hop UAV networks. However, as the number of agents increases, the environment becomes unstable, which poses a challenge to value-based methods. In addition, the above methods [34], [35], [36] are all applicable to small-scale UAV swarms.

Policy-based methods directly learn the optimal policy, making this direct optimization approach especially effective in unstable environments [37]. Chen et al. [20] studied the joint optimization problem of UAV relay stations and ground users, and regarded it as a hybrid cooperative game problem with two kinds of intelligence modeling. The authors proposed a UAV trajectory planning algorithm based on the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) model to maximize fairness throughput. Addressing the security issues of UAV base station communication, Zhang et al. [38] proposed a cooperative interference method, leveraging UAV jamming UAVs to assist UAV transmitters in defending against ground entity attacks. Furthermore, a cooperative trajectory planning algorithm based on the MADDPG model is proposed to jointly optimize the flight trajectories and transmission power of both types of UAVs. In the aforementioned works [20], [34], [35], [36], [38], the scale of UAVs is relatively small. Due to the centralized training structure of CTDE, issues such as dimensionality catastrophe and environmental stability degradation arise when the scale of UAVs increases.

III. Heterogeneous Multi-UAV System Model

As shown in Fig. 1, the paper addresses an emergency search and rescue scenario characterized by complex terrain, multiple obstacles, and lack of communication coverage. A leader-follower multi-UAV cooperative architecture is employed. The multi-UAV system comprises F fixed-wing UAVs equipped with long-distance communication capabilities, serving as relays for multi-rotor UAVs. Additionally, there are N multi-rotor UAVs equipped with Received Signal Strength Indication (RSSI) detection devices capable of detecting the strength of target signal sources. The set of multi-UAVs is denoted as $\mathcal{N} = \{1, 2, \ldots, N + F\}$, with their coordinates at timeslot t denoted as

$$p_i(t) = [x_i(t), y_i(t), z_i(t)] \in \mathbb{R}^{3\times 1}, i \in \mathcal{N}. \tag{1}$$

The K targets to be searched on the ground can be represented by a set denoted as $\mathcal{K} = \{1, 2, \ldots, K\}$. Their relative coordinates are specified by

$$p_k(t) = [x_k(t), y_k(t)] \in \mathbb{R}^{2\times 1}, k \in \mathcal{K}. \tag{2}$$

The system can perceive the signal strength of ground targets only within the sensing range of multi-rotor UAVs to determine the distance from the target.

A. Communication Model

Heterogeneous UAVs employ clustering technology. In this swarm, fixed-wing UAVs act as central nodes and connect to multi-rotor UAVs in a star configuration. Meanwhile, the fixed-wing UAVs form a mesh network among themselves. As discussed in [39], communication between multi-rotor UAVs and fixed-wing UAVs can be illustrated by a probabilistic path loss model, where the path loss is calculated as the probability-weighted sum of Line-of-Sight (LoS) and Non-Line-of-Sight (NLoS) links, and the average path loss for multi-UAV is calculated as

$$L_{\text{avg}}^{i,j}(t) = P_{\text{LoS}}^{i,j}(t) \times L_{\text{LoS}}^{i,j} + \left(1 - P_{\text{LoS}}^{i,j}(t)\right) \times L_{\text{NLoS}}^{i,j}. \quad (3)$$

Based on the lightweight Sigmoid model verified by [40], [41], the LoS path probability $P_{\text{LoS}}^{i,j}(t)$ is calculated as

$$P_{\text{LoS}}^{i,j}(t) = \frac{1}{1 + a \cdot e^{-b(\theta_{i,j} - a)}}, \quad (4)$$

where a is the environmental shadowing factor, which reflects obstacle density and spatial distribution characteristics, and b denotes the slope factor of LoS probability transition, which indicates the height and spatial sharpness of obstacles. $\theta_{i,j}$ represents the elevation angle between UAV i and UAV j. However, the $P_{\text{LoS}}^{i,j}(t)$ among fixed-wing UAVs is 1 since they operate above building heights, eliminating obstructions between communication links. The $L_{\text{LoS}}^{i,j}$ is the LoS path loss, $L_{\text{NLoS}}^{i,j}$ is the NLoS path loss, and are calculated as

$$L_{\text{LoS}}^{i,j} = G^{i,j}(t) + \nu_{\text{LoS}}, \quad (5a)$$

$$L_{\text{NLoS}}^{i,j} = G^{i,j}(t) + \nu_{\text{NLoS}}, \quad (5b)$$

where ν_{LoS} and ν_{NLoS} are the mean additional losses for LoS and NLoS, respectively. $G^{i,j}(t)$ is free space path loss. The Signal-to-Interference-plus-Noise Ratio (SINR) is calculated as

$$\gamma_{\text{A2A}}^{i,j}(t) = \frac{P_{tx,u}^{j} \cdot \frac{1}{L_{\text{avg}}^{i,j}(t)}}{\sum_{z=1, z \neq i}^{|\mathcal{L}|} I_{i,z} + \upsilon_{A2A}^2}, \quad (6)$$

where $P_{tx,u}^{j}$ is the transmission power of UAV j. When $u = fix$, $P_{tx,u}^{j}$ denotes transmission power of the fixed-wing UAV. $|\mathcal{L}|$ is the total number of UAVs that communicate with UAV i, and $I_{i,z}$ is interference power from UAV z, υ_{A2A}^2 is Gaussian noise. The communication rate from UAV j to UAV i is denoted as $R_{\text{A2A}}^{i,j}$ and is calculated as

$$R_{\text{A2A}}^{i,j}(t) = B \log_2 \left(1 + \gamma_{\text{A2A}}^{i,j}(t)\right). \quad (7)$$

where B is bandwidth.

B. Dynamic Model of UAVs

To reduce model complexity and focus more on trajectory optimization, this paper simplifies the dynamics model of fixed-wing UAVs by neglecting pitch and roll attitudes, and retaining yaw variation, i.e., the rotation of Z-axis of the aircraft body coordinate. The displacement kinematics equation is denoted as

$$\dot{p} = R(\gamma)v, \quad v \in \mathbb{R}^{3\times 1}, \quad \gamma \in \mathbb{R}^{1\times 1}, \quad (8)$$

where, $\dot{p}$ denotes position offset, v denotes the UAV velocity vector, R denotes the rotation matrix, and γ denotes the yaw rotation angle. The simplified Newton-Euler motion equation for fixed-wing UAVs is

$$m_f \dot{v} + \frac{1}{2}\rho v^2 = F_{\text{air}}, \quad F_{\text{air}} \in \mathbb{R}^{3\times 1}, \quad (9)$$

$$I_{\text{yaw}}\dot{\gamma} + I_{\text{yaw}}\gamma^2 = R, \quad (10)$$

where F_{air} denotes the propulsion force output by the UAV propellers, m_f denotes the mass of the fixed-wing UAV, ρ denotes the air resistance coefficient, I_{yaw} is the moment of inertia about the yaw axis, and R represents the torque generated by the directional rudder.

For multi-rotor UAVs, without considering attitude changes, the dynamics model is denoted as

$$\dot{p} = v, \quad v \in \mathbb{R}^{3\times 1}, \quad (11)$$

$$m_u \dot{v} + \frac{1}{2}\rho v^2 = F_{\text{air}}, \quad F_{\text{air}} \in \mathbb{R}^{3\times 1}, \quad (12)$$

where m_u denotes the mass of the multi-rotor UAV.

C. Energy Consumption Model

The energy consumption of multi-rotor UAV consists of two main components: flight energy consumption and communication energy consumption. The flight energy consumption of multi-rotor UAV is mainly positively correlated with flight speed v and acceleration a. The formula for flight energy consumption is denoted as

$$P_{\text{dyn}}(v, a) \approx P_0 + \mu v^2 + d a v^3, \quad (13)$$

where P_0 is the energy consumption of the UAV to offset its own gravity, μ is the coefficient of blade resistance energy consumption, and d is the coefficient of airframe resistance energy consumption.

Communication-related energy consumption includes the energy consumption of communication circuits, signal processing, and computation. This part of energy consumption accounts for a very small proportion of the overall UAV energy consumption. To simplify the system model complexity, this paper sets the communication energy consumption as a constant value P_{com}. Finally, the total mission energy consumption of multi-rotor UAV combining communication energy consumption and flight energy consumption is denoted as

$$E_i(T) = \int_0^T \left(P_{\text{dyn}}(v(t), a(t)) + P_{com}\right) dt. \quad (14)$$

Due to the large payload and long endurance of the fixed-wing UAVs, their flight time is much larger than that of the multi-rotor UAVs, and thus the optimization of energy consumption in fixed-wing UAVs is not necessary for the SAR mission. Therefore, this paper does not consider energy consumption optimization for fixed-wing UAVs.

D. Problem Formulation

This paper considers the optimization objectives of target detection quantity and mission energy consumption to maximize mission energy efficiency. Multi-rotor UAVs need to quickly complete the search of task areas following fixed-wing UAVs, while maintaining effective communication among all UAV nodes at a certain network rate. Multi-rotor UAVs need to effectively navigate obstacles in complex post-disaster terrain and optimize their trajectories and maneuvers to reduce energy consumption. To address the above concerns, this paper establishes a multi-constraint multi-objective optimization model to maximize mission energy efficiency by optimizing UAV trajectories. Due to the much longer endurance of fixed-wing UAVs compared to multi-rotor UAVs, the energy consumption of fixed-wing UAVs will not be considered in the model. The problem is modeled as follows.

$$\begin{aligned} &\mathbf{P1}: \max_{\{p_i(t), v_i(t)\}} \left\{ \frac{\sum_{i=1}^{N} N_i(T)}{\sum_{i=1}^{N} E_i(T)} \right\} \\ &\text{s.t.:} \\ &\quad \text{C1: } E_i(T) > e_{\text{safe}}, \\ &\quad \text{C2: } R_{\text{A2A}}^{i,j}(t) > R_{\min}, \\ &\quad \text{C3: } p_i(t) \in \Omega_i, p_i(t) \notin \Omega_{\text{obs}}, \\ &\quad \text{C4: } p_i(t), p_j(t) \in \Omega_{\text{task}}, \\ &\quad \text{C5: } v_i(t) < V_{\max}, v_{\text{fix}}(t) \geq V_{\min}, \end{aligned} \tag{15}$$

where $N_i(T)$ denotes the number of targets detected by UAV i during the entire mission, $E_i(T)$ denotes the percentage of battery consumption for UAV i during the entire mission, and the ratio of the $N_i(T)$ to $E_i(T)$ is the energy efficiency of the multi-objective search mission. e_{safe} denotes the minimum safe battery capacity of the UAV. Constraint C1 ensures safe battery consumption for UAVs to reserve enough battery for return. Constraint C2 ensures that the communication rate among network nodes reaches a certain value to meet stable communication among multi-rotor UAVs and the fixed-wing UAV. Herein, $R_{\min}$ is the minimum communication rate established among UAVs. Constraint C3 specifies that Ω_i is the safety domain for UAV i, where Ω_{obs} is the obstacle domain, and UAVs are restricted from flying into this area. p_j denotes coordinate of the fixed-wing UAV. Constraint C4 specifies that Ω_{task} is the restricted area for tasks. Constraint C5 imposes velocity constraints, where $V_{\max}$ is the maximum flight speed constraint, $v_{\text{fix}}(t)$ is the fixed-wing UAV speed, and $V_{\min}$ is the minimum flight speed constraint to avoid stalling.

IV. GATAC Algorithm

In this paper, the multi-UAV cooperative trajectory optimization problem is transformed into a Markov Games problem, which is an extension of Markov Decision Processes (MDPs) designed for multi-agent systems. We propose the GATAC algorithm based on CTDE and MARL to optimize problem **P1**. The structure of the algorithm is illustrated in Fig. 2. The GATAC algorithm effectively tackles the challenge of dimensionality explosion that arises as the number of agents increases.

A. Problems Transformation

First, the total task time T is divided into M_c timeslots, where $T = M_c \cdot \delta_t$. Within timeslot δ_t, the actions, cooperation

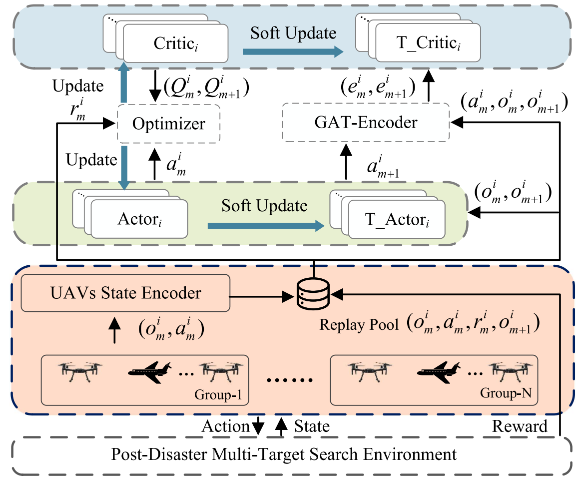


Fig. 2. GATAC algorithm architecture.

strategies, and network states of the multiple UAVs remain approximately constant. The problem **P1** is discretized into

$$\begin{aligned} &\mathbf{P2}: \max_{\{p_i(m), v_i(m)\}} \left\{ \frac{\sum_{i=1}^{N} N_i(M_c \cdot \delta_t)}{\sum_{i=1}^{N} E_i(M_c \cdot \delta_t)} \right\} \\ &\text{s.t.:} \\ &\quad \text{C1: } E_i(M_c \cdot \delta_t) > e_{\text{safe}}, \\ &\quad \text{C2: } R_{\text{A2A}}^{i,j}(m) > R_{\min}, \\ &\quad \text{C3: } P_i(m) \in \Omega_i, P_i(m) \notin \Omega_{\text{obs}}, \\ &\quad \text{C4: } P_i(m), P_j(m) \in \Omega_{\text{task}}, \\ &\quad \text{C5: } V_i(m) < V_{\max}, V_{\text{fix}}(m) \geq V_{\min}. \end{aligned} \tag{16}$$

The three key parameters $\{S, A, R\}$ of the MDPs are demonstrated as follows.

S is the state space $s_{u,m}^i \in S$, and $u \in \{mul, fix\}$, where mul denotes multi-rotor UAV and fix denotes fixed-wing UAV, and $m \in \{1, 2, \ldots, M_c\}$. This paper considers two types of UAVs, and two state spaces are designed. When $u = mul$, the state of multi-rotor UAV i at timeslot m is denoted as

$$s_{u,m}^i = \{p_m^i, v_m^i, e_m^i, n_m^i\}, \forall i \in \mathcal{N}. \tag{17}$$

The local observation environment state of multi-rotor UAV i needs to obtain information from other UAVs for cooperation and is denoted as

$$o_{u,m}^i = \{s_{u,m}^i, \{d_m^{i,j}, d_m^{i,k}, p_m^j, e_m^j, n_m^j\}\}, \forall i, j \in \mathcal{N}, \forall k \in \mathcal{K}. \tag{18}$$

When $u = fix$, the state of fixed-wing UAV i at timeslot m is denoted as

$$s_{u,m}^i = \{p_m^i, v_m^i, \psi_m^i\}, \forall i \in \mathcal{N}, \tag{19}$$

and the local observation environment state of fixed-wing UAV i is denoted

$$o_{u,m}^i = \{s_{u,m}^i, \{d_m^{i,k}, p_m^j, n_m^j\}\}, \forall i, j \in \mathcal{N}, \forall k \in \mathcal{K}, \tag{20}$$

where p_m^i is the position of UAV i, v_m^i is the speed of the UAV i, e_m^i is the energy level of the multi-rotor UAV i, n_m^i is the network connection state of the UAV i, $d_m^{i,j}$ and $d_m^{i,k}$ are the distances to other cooperative UAVs and target points respectively, and ψ_m^i is the heading angle of fixed-wing UAV i.

A is the action space, and $a_{u,m}^{i} \in A$. When $u = mul$, $a_{u,m}^{i}$ represents the actions of the multi-rotor UAV i at timeslot m and is denoted as $a_{u,m}^{i} = \{F_i(m), \theta_i(m)\}$, where $F_i(m)$ is the force scalar. When $u = fix$, $a_{u,m}^{i}$ represents the actions of the fixed-rotor UAV i at timeslot m and is denoted as $a_{u,m}^{i} = \{\dot{\gamma}_i(m), F_i(m)\}$, where $\dot{\gamma}_i(m)$ is the heading rotational angle acceleration scalar in both clockwise and counterclockwise directions, and $F_i(m)$ is the thrust scalar output by fixed-wing UAV i.

R is the reward function of the model, which is essential for completing training in MARL and directly affects the performance of the model. In this paper, two reward functions are set for the multi-rotor UAV and fixed-wing UAV, respectively, which include communication, energy, safety, and search task reward functions.

1) Communication Reward: The communication reward guides multi-rotor UAV to maintain effective communication with fixed-wing UAVs. The communication reward $r_{\text{com}}^{i,m}$ for multi-rotor UAV i at timeslot m is denoted as

$$r_{\text{com}}^{i,m} = \begin{cases} -r_c, & R_{\text{A2A}}^{i,j}(m) < R_{\min} \\ \frac{r_c}{d_m^{i,j} + \Delta d}, & R_{\max} > R_{\text{A2A}}^{i,j}(m) > R_{\min} \\ r_c, & R_{\text{A2A}}^{i,j}(m) > R_{\max} \end{cases}, \quad (21)$$

where r_c is the maximum communication reward, $d_m^{i,j}$ is the distance to establish communication at timeslot m, Δd prevents division by zero, $R_{\text{A2A}}^{i,j}(m)$ is the air-to-air network communication rate, $R_{\min}$ is the minimum communication rate, and $R_{\max}$ is the maximum communication rate. When the communication rate has not reached $R_{\max}$, the reward is negatively correlated with distance. When the rate drops below $R_{\min}$, the UAV receives a negative reward $-r_c$ as a penalty.

2) Energy Consumption: The energy consumption reward for multi-rotor UAV i at timeslot m is denoted as

$$r_{\text{power}}^{i,m} = \begin{cases} \tau e_m^i, & e_m^i > e_{\text{safe}} \\ 0, & e_m^i \le e_{\text{safe}} \end{cases}, \quad (22)$$

where e_m^i is the remaining power of UAV i at timeslot m, and e_{safe} is the safe power threshold. Since the value of e_m^i is very larger than other reward, energy consumption reward scaling factor τ ($\tau \in [0, 1]$) is required to reduce the proportion of power consumption in the total reward.

3) Safety Reward: The safety reward for multi-rotor UAV i at timeslot m is denoted as

$$r_{\text{safe}}^{i,m} = \begin{cases} \frac{-\eta}{d_m^{i,j} + \Delta d}, & d_m^{i,j} \le D_{\text{safe}} \\ 0, & d_m^{i,j} > D_{\text{safe}} \end{cases}, \quad (23)$$

where η is the safety reward coefficient, $d_m^{i,j}$ is the distance to other UAVs at timeslot m, and D_{safe} is the safety distance threshold.

4) Search Task Reward: The search task reward for multi-rotor UAV i at timeslot m is denoted as

$$r_{\text{task-m}}^{i,m} = \begin{cases} \frac{\zeta}{d_m^{i,k} + \Delta d}, & D_{\text{found}} < d_m^{i,k} \le D_{\text{detect}} \\ \zeta, & d_m^{i,k} < D_{\text{found}} \end{cases}, \quad (24)$$

where D_{detect} is the maximum detection distance, D_{found} is the distance for detection and confirmation, ζ is the task reward coefficient, and $d_m^{i,k}$ is the distance to the search target at timeslot m. The detect reward for fixed-wing UAV i at timeslot m is denoted as

$$r_{\text{task-f}}^{i,m} = \begin{cases} \frac{\zeta}{d_m^{i,k} + \Delta d}, & d_m^{i,k} \le D_{\text{detect-f}} \\ 0, & d_m^{i,k} > D_{\text{detect-f}} \end{cases}, \quad (25)$$

where $D_{\text{detect-f}}$ is the maximum detection distance of the fixed-wing UAV.

5) Total Reward: In summary, the total reward at timeslot m received by multi-rotor UAV i is denoted as

$$r_{\text{m}}^{i,m} = r_{\text{com}}^{i,m} + r_{\text{power}}^{i,m} + r_{\text{safe}}^{i,m} + r_{\text{task-m}}^{i,m}, \quad (26)$$

due to the difference with the task of the multi-rotor UAV, the fixed-wing UAV i receives a reward at timeslot m as

$$r_{\text{f}}^{i,m} = r_{\text{safe}}^{i,m} + r_{\text{task-f}}^{i,m}. \quad (27)$$

6) Reward Coefficient Analysis: In the context of multi-rotor UAV formation tasks, it is crucial to consider a range of factors that influence overall performance, including balanced exploration, communication efficiency, and safety. Specifically, the main goal of multi-rotor UAVs is to discover as many ground targets as possible. Upon successfully discovering a target, a UAV is awarded a good reward. To address the risks associated with collisions and chain breaks among UAVs, greater penalties or rewards are applied based on the distance between UAVs, thereby ensuring safety within the formation. Furthermore, a small reward will be given to the multi-rotor UAV based on remaining power of the UAV itself.

B. MARL Framework

This paper constructs an MARL framework based on policy gradient, utilizing a CTDE structure. Each UAV possesses a state set S_m consisting of observations $\{o_1, o_2, \ldots, o_{N+M}\}$, where o_i represents the local information observed by UAV i, and an action set $A_m = \{a_1, a_2, \ldots, a_{N+M}\}$. The expected cumulative reward for UAV i is denoted as

$$J_m^i(\pi_i) = \mathbb{E}_{A_m \sim \pi_i}\left[\gamma R_m^i(S_m, A_m)\right], \quad (28)$$

where $R_m^i(S_m, A_m)$ denotes the reward obtained when all agents take actions A_m in state S_m, and π_i is the policy function learned by each agent (Actor network), and $\gamma \in [0, 1]$ represents the reward discount factor. The policy gradient aims to estimate the gradient of the expected return of an agent with respect to its policy parameters. After incorporating action entropy, the policy gradient formula is

$$\nabla_{\varphi_i} J_m(\pi_{\varphi_i}) = \mathbb{E}_{a_m^i \sim \pi_{\varphi_i}}\left[\nabla_{\omega_i} \log(\pi_{\varphi_i}(a_m^i | o_m^i)) y_m^i\right], \quad (29)$$

$$y_m^i = Q_{\omega_i}(S_m, A_m) - \alpha \log\left(\pi_{\varphi_i}\left(a_m^i | o_m^i\right)\right), \quad (30)$$

where π_{φ_i} denotes the Actor network of UAV i and Q_{ω_i} denotes the Critic network. Both the Actor network and the Critic network for UAV i are constructed using artificial neural networks. This paper adopts a CTDE architecture, where Critic networks share a common loss function to jointly update parameters for minimizing the error. α is the coefficient for action entropy, which reflects the degree of exploration of the agents. φ_i and

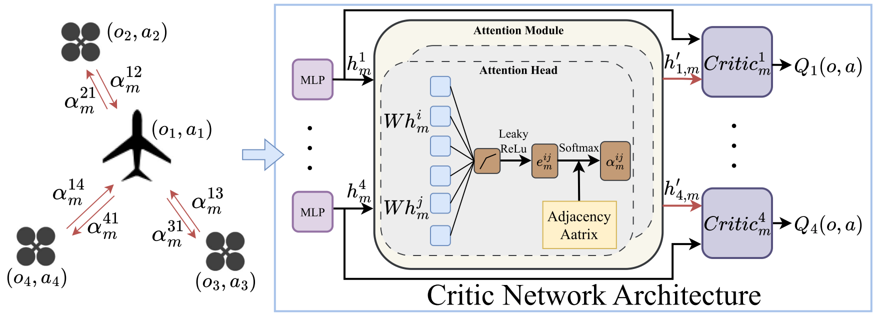


Fig. 3. Critic network architecture of the GATAC algorithm and GAT computation process.

ω_i are the parameters for the Actor and Critic networks, respectively. B denotes the experience buffer that stores local state observations o_m^i, selected actions a_m^i, subsequent state o_{m+1}^i, and corresponding rewards r_m^i. The loss function of the Q_{ω_i} network is denoted as

$$\mathcal{L}_Q\left(\varphi_i\right)=\sum_{i=1}^{N}\Big(Q_{\omega_i}\left(S_m,A_m\right)-y_m^i\Big)^2, \tag{31}$$

where y_m^i is the training target and is calculated as

$$y_m^i=r_m^i+\gamma Q_{\overline{\omega_i}}\left(S_m,A_m\right)-\alpha\log\left(\pi_{\overline{\varphi_i}}\left(a_{m+1}^i \mid o_{m+1}^i\right)\right), \tag{32}$$

where $Q_{\overline{\omega_i}}$ is the target Critic network, $\pi_{\overline{\varphi_i}}$ is the target Actor network, and $\overline{\varphi_i}$ and $\overline{\omega_i}$ are the parameters of the target Actor and Critic networks, respectively. The update formula for parameter $\overline{\varphi_i}$ and $\overline{\omega_i}$ is given as follows

$$\overline{\varphi_i}=\varepsilon\varphi_i+(1-\varepsilon)\overline{\varphi_i}, \tag{33}$$

$$\overline{\omega_i}=\varepsilon\omega_i+(1-\varepsilon)\overline{\omega_i},\varepsilon\in[0,1], \tag{34}$$

where ε is the soft update coefficient.

C. Graph Attention Encoder

In the CTDE architecture, effective training of Critic networks is crucial to guide the training of the policy network of each UAV. However, it is evident that as the number of UAVs and the state space increase, the dimensionality of Critic network inputs grows exponentially. This leads to the problem of the curse of dimensionality, and excessive irrelevant information from other UAVs can also cause interference, increasing the non-stationarity of the environment. Based on the tasks and functionalities of UAVs, the establishment of a cooperative Graph Attention Network (GAT) can effectively address these issues. This section primarily introduces how to integrate a GAT into the policy gradient-based MARL algorithm.

First, as shown in Fig. 3, this paper constructs a cooperative relationship graph, where each UAV is represented as a node in the graph, and the connectivity among UAVs are represented as edges. This graphical approach sparsifies the cooperation of UAVs according to the needs of the task, which is equivalent to determining a fuzzy guidance of the collaboration. Additionally, structure of such graph makes the cooperative relationships among UAVs more intuitive and facilitates the understanding of the network of UAV relationships. Then, the importance weights of cooperation are calculated using the graph attention model

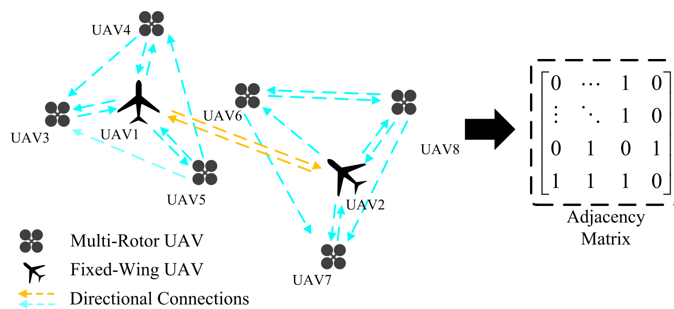


Fig. 4. UAV cooperation graph structure.

of Fig. 3. Final, a fixed cooperation model is formed through training. By incorporating the graph attention model into the Critic network, an efficient information aggregation encoder is effectively introduced, which greatly improves the training of MARL in complex tasks.

Moreover, due to the inclusion of the graph structure, the Critic network evaluates a fixed number of UAVs regarding state and action, thereby avoiding the curse of dimensionality as the total system scale increases, this design makes the model more efficient and stable when dealing with large-scale multi-rotor systems. Additionally, this clustering method not only improves the efficiency of task execution, but also enhances the collaborative ability between different types of UAVs, enabling the entire system to better respond to complex task requirements and dynamic environmental changes. The following section will detail the GAT encoder.

As shown in Fig. 3, the cooperative interaction among UAVs at timeslot m is formally modeled as a graph $\mathcal{G}=\{\mathcal{V},\mathcal{A}\}$, where $\mathcal{V}$ denotes the set of nodes and $\mathcal{A}$ represents the adjacency matrix encoding inter-node connections. The node set is defined as $\mathcal{V}=\{(a_m^i,o_m^i)\mid i\in\mathcal{N}\}$, where each node corresponds to the action a_m^i and observation state o_m^i of UAV i. These node attributes are subsequently encoded into latent feature representations h_m^i through a MLP encoder.

$$h_m^i=f_{\text{encoder}}(a_m^i,o_m^i), \tag{35}$$

where f_{encoder} is the MLP encoder. The adjacency matrix $\mathcal{A}$ defines the connections among all UAV nodes, and its construction is shown in Fig. 4. In the initial state, multi-rotor UAVs are connected only to other UAVs within the same formation, while fixed-wing UAVs are connected to other UAVs within the same formation and other fixed-wing UAVs. However, during the mission, communication limitations among UAVs can cause connections involving multi-rotor UAVs to be lost. This may result in information transfer failures. Therefore, the adjacency matrix $\mathcal{A}$ requires real-time updates specifically for connections between UAVs that belong to the same formation. For connection $\mathcal{A}_{i,j}$ from UAV j to UAV i, where both UAV i and UAV j are in the same formation, the connection status is determined by

$$\mathcal{A}_{i,j} = \begin{cases} 1, & \text{if } R^{i,j}_{\text{A2A}} > R_{\min} \\ 0, & \text{otherwise} \end{cases}, \tag{36}$$

where $\mathcal{A}_{i,j} = 1$ indicates a communication link from UAV j to UAV i. Conversely, $\mathcal{A}_{i,j} = 0$ signifies no direct connection. Therefore, the set of cooperative UAV collection for node i is defined as $\mathcal{L} = \{1, 2, \ldots, L\}$ based on $\mathcal{A}$.

As shown in the attention module of Fig. 3, the similarity coefficients between UAV i and each adjacent UAV are calculated sequentially. The similarity coefficient $e^{i,j}_m$ between i and j is calculated as

$$e^{i,j}_m = \sigma_L \left(a^{\mathrm{T}} \left[W h^i_m \parallel W h^j_m \right] \right), \tag{37}$$

where σ_L denotes the LeakyReLU activation function, $\parallel$ denotes the concatenation operation, and W is the linear transformation weight matrix. The attention coefficient $\alpha^{i,j}_m$ between UAV j and UAV i is computed as

$$\alpha^{i,j}_m = \frac{\exp(e^{i,j}_m)}{\sum_{k \in L} \exp(e^{i,k}_m)}. \tag{38}$$

Then, all the node information connected to node i is aggregated by

$$h'_{i,m} = \sigma_R \left(\sum_{j=0}^{|\mathcal{L}|} \alpha^{i,j}_m W h^j_m \right), \tag{39}$$

where h^j_m corresponds to the feature vector of UAV j, and σ_R is the ReLU activation function. Finally, the aggregated feature $h'_{i,m}$ and node feature h^i_m are input into the Critic network to calculate Q_{w_i}.

D. Multi-UAVs State Observation Optimization

This paper adopts a CTED architecture for MARL, where each UAV is assigned an Actor network to output actions. The Actor network's input consists primarily of two parts: one part is the self-state information, and the other part is the observed information from other UAV nodes and search targets. If all the UAV states information are input into the Actor network, the network may be interfered with information from UAVs unrelate to itself during computation and would not meet practical system requirements. Therefore, this paper proposes a cooperation importance index to sort UAV state information and remove UAV information with lower importance. The cooperation importance index is mainly composed of distance, battery level, and the number of detected signal sources within the detection range and is formulated as

$$\eta^{i,j}_m = w_{ud} \frac{d^{i,j}_m}{D_{\max}} + w_e \frac{e^j_m}{E_{\max}} + w_n \frac{n^j_m}{N_{\max}}, \tag{40}$$

where $\eta^{i,j}_m$ represents the importance of UAV j to UAV i, $d^{i,j}_m$ is the distance between UAV i and UAV j, e^j_m is the remaining battery level of UAV j, n^j_m is the number of target signals detected by UAV j. w_{ud}, w_e, and w_n are the weights representing the importance of distance, battery level, and the number of detected signal sources within the detection range, respectively. $D_{\max}$, $E_{\max}$, and $N_{\max}$ are the maximum possible values of distance, battery level, and signal source count, respectively. Additionally, as the mission progresses, the importance of UAVs will change, leading to changes in the concatenation order of information vectors of cooperating UAVs. Therefore, this paper also includes the UAV number as part of the state information.

Furthermore, when UAVs conduct multi-target searches, too much acquired target information will increase the input vector dimensionality of the UAV Actor network. This is primarily because such data can significantly increase the dimensionality of the input vectors required by the UAV Actor network, potentially leading to inefficiencies and complexities in processing and decision-making. Therefore, a target search importance index is also proposed to optimize target information observation. The target search importance index is mainly composed of target distance and the waiting time for targets discovered and is formulated as

$$\beta^{i,k}_m = w_{td} \frac{d^{i,k}_m}{D_{\max}} + w_t \frac{t^k_m}{T_{\max}}, \tag{41}$$

where $\beta^{i,k}_m$ denotes the importance of target k to UAV i, and w_{td} and w_t are the weights representing the importance of target distance and waiting time, respectively. $D_{\max}$ and $T_{\max}$ are the maximum possible values of distance and waiting time, used for normalization. Additionally, once a target is discovered, it is no longer included in the importance index calculation and is masked from observation. When the remaining targets are fewer than the predefined target number of the UAV Actor network, they are filled with masks. The above method of multi-UAV state observation optimization is shown in Fig. 5.

V. Training Process

This section provides a detailed description of the training process of the GATAC algorithm, which consists of three main stages: initialization, sampling, and parameter updating. The specific training process is outlined in Algorithm 1 as follows.

Initialization Stage (Line 1–3): At the beginning of the training phase, the Actor and Critic network parameters are initialized for each UAV, and copy these parameters to the target Actor and Critic networks. The GATAC algorithm instantiates the experience replay buffer B and imports the adjacency matrix of the UAV cooperative graph. Additionally, the states of all UAVs and the environment are initialized.

Sampling Stage (Lines 5–13): In this stage, UAVs input their observed local states $o^i_{u,m}$ into the Actor network. Subsequently, the Actor network outputs the probability values for each action, which are then combined with random action probabilities to determine actions $a^i_{u,m}$. After executing action $a^i_{u,m}$, UAVs transfer their own states to the next timeslot $o^i_{u,m+1}$, and rewards r^i_m are computed by using the reward function. The experiences

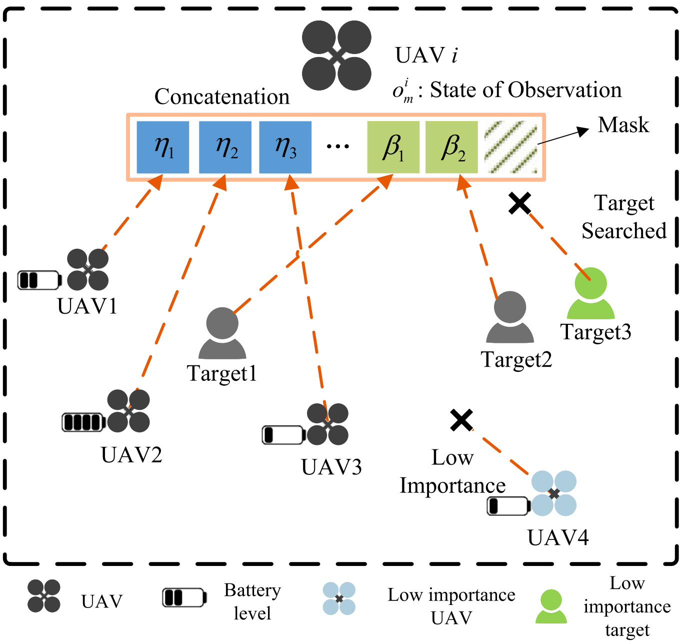


Fig. 5. Optimization method for multi-UAV state observation.

of all UAV $(o_{u,m}, a_{u,m}, r_m, o_{u,m+1})$ obtained after execution are stored in the experience buffer B for updating network parameters in the next timeslot.

Parameter Updating Stage (Lines 15–18): In this stage, a batch of data is randomly selected from experience buffer B and reward normalization is performed. The experience data of the UAVs is then aggregated by a graph attention encoder. Finally, the Actor and Critic network parameters are updated for each UAV. After each parameter update, the target Actor and Critic network parameters are synchronized to the Actor and Critic networks through soft updates.

The computational complexity of the GATAC algorithm is divided into an execution phase and a training phase. Only Actor network participate in the computation during the execution phase, and the computational complexity of the execution phase is calculated as $O(|\mathcal{N}| \times \sum_{j=0}^{L-1}(f_j \times f_{j+1}))$, where L represents the number of layers in Actor network, and f_j represents the dimension of the j-th fully connected layer. During the training phase, both Actor and Critic will participate in the training. The complexity analysis of the Critic and Target-Critic networks is calculated similarly to Actor networks and is calculated as $O(|\mathcal{N}| \times \sum_{h=0}^{J-1}(f_h \times f_{h+1}))$, where J represents the number of Critic and Target-Critic networks layers, and f_h represents the dimension of the h-th fully connected layer. The complexity calculation of the attention module is calculated as $O(\sum_{i=0}^{Q-1}(f_i \times f_{i+1}))$. In summary, the computational complexity during the training phase is calculated as $O(2 \times |\mathcal{N}| \times (\sum_{j=0}^{L-1}(f_j \times f_{j+1}) + \sum_{h=0}^{J-1}(f_h \times f_{h+1})) + \sum_{i=0}^{Q-1}(f_i \times f_{i+1}))$.

VI. Experiment and Result Analysis

In this section, we introduce a simulation scenario and design experiments to compare the training reward values, average numbers of searched targets, and average energy consumptions

Algorithm 1: GATAC Algorithm Training Process.

1: **Input**: Observation state of multi-rotor UAV and fixed-wing UAV observation state: $o_{u,m}^i$.
2: **Output**: Multi-rotor UAV actions and fixed-wing UAV actions: $a_{u,m}^i$.
3: $\triangleright$ **Initialization**:
Initialize the critic and actor networks parameters ϕ_i and ω_i;
4: **for** each episode in E **do**
5: Initialize the state of UAV i and the environment;
6: **for** each step m in M **do**
7: $\triangleright$ **Sampling**
8: **for** UAV i **in** $\mathcal{N}$ **do**
9: Select action $a_m^i = \pi_{\varphi_i}(a_m^i | o_m^i) + \eta$;
10: **end for**
11: UAVs execute their actions $a_m = (a_m^1, \ldots, a_m^N)$;
12: Obtain next state s_{m+1} and reward $r_m = (r_m^1, \ldots, r_m^N)$;
13: Store obtained $(o_{u,m}, a_{u,m}, r_{u,m}, o_{u,m+1})$ in experience pool B;
14: $\triangleright$ **Parameter Update**
15: **for** UAV i in $\mathcal{N}$ **do**
16: Randomly select a batch of data from experience pool B;
17: Update weights φ_i and ω_i according to equation (33) and (34);
18: **end for**
19: **end for**
20: **end for**

of UAVs between the GATAC algorithm and four other MARL algorithms.

MADDPG [42]: The algorithm is an MARL algorithm based on the CTDE architecture. It's widely applicable to cooperative tasks among multiple agents, effectively addressing training challenges in non-stationary environments.

Multi-Agent Twin Delayed Deep Deterministic Policy Gradient (MATD3) [43]: It addresses the overestimation of Q-values in MADDPG using clipped double Q-learning and target policy smoothing, thereby achieving more robust cooperative strategies.

Multi-Agent Soft Actor-Critic (MASAC) [44]: The MASAC algorithm is widely used in multi-agent cooperation tasks. It utilizes entropy regularization to facilitate policy exploration, which helps to find more robust cooperative strategies in multi-agent environments. The GATAC algorithm also leverages entropy regularization to improve policy exploration.

Self Attention-GATAC (SA-GATAC): This algorithm retains the framework of multi-UAV cooperative graphs but changes the way node information is aggregated to a self-attention model, aiming to verify performance differences between the two attention calculation models.

A. Simulation Design

As shown in Fig. 6, this paper constructs a heterogeneous multi-UAV post-disaster target search scenario based on the

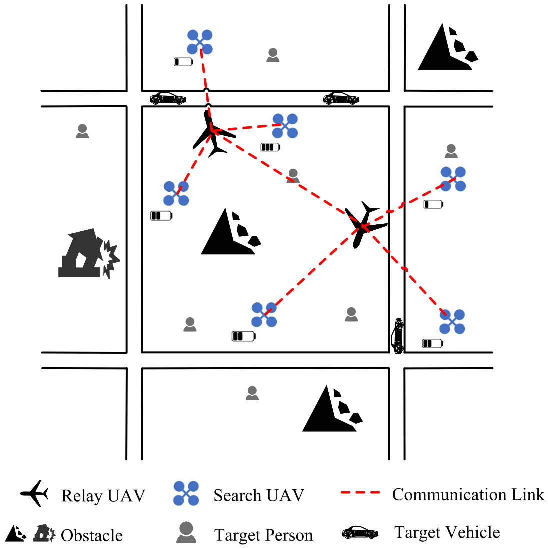


Fig. 6. Simulation scenario of post-disaster search and rescue with heterogeneous multi-UAVs.

TABLE I
EXPERIMENTAL PARAMETERS

Parameter	Value
Fixed-Wing UAV weight (m_{f})	10kg
Multi-rotor UAV weight (m_{u})	1kg
Communication power (P_{com})	5W
Minimum communication rate ($R_{\min}$)	1Mbps
Max flight speed (Fixed-wing) ($V_{\max}$)	40m/s
Min flight speed (Fixed-wing) ($V_{\min}$)	10m/s
Max acceleration ($a_{\max}$)	8m/s^2
Max flight speed (Multi-rotor) ($V_{\max}$)	10m/s
Safety battery level (e_{safe})	10%
Energy consumption reward scaling factor (τ)	0.2
Training rounds (E)	50,000
Episode length (M)	50
Learning rate (α)	0.001
Reward discount factor (γ)	0.99
Soft update coefficient (ε)	0.99

OpenAI multi-particle environment. The simulation domain is a 5 km $\times$ 5 km rectangle populated by multiple search targets and buildings, with their locations drawn from a uniform random distribution over the area. To simplify geometric representation, the buildings are modeled as circles.

UAV models include fixed-wing and multi-rotor types based on UAV dynamics formulas. Fixed-wing UAVs serve as mobile relay stations providing continuous network coverage for multi-rotor UAVs. They are additionally equipped with signal detection devices to measure signal strengths within their coverage area. Multi-rotor UAVs are equipped with cameras for multi-target search and confirmation in low-altitude complex environments. The simulation platform is based on Ubuntu 20.04.4, utilizing PyTorch 1.7 and Python 3.8 software. Hardware includes Intel Xeon (R) Gold 5220R CPUs and NVIDIA RTX 6000 GPUs. In practical applications, each UAV can carry independent edge computing devices, enabling a decentralized cooperative system. Experimental parameters are detailed in Table I.

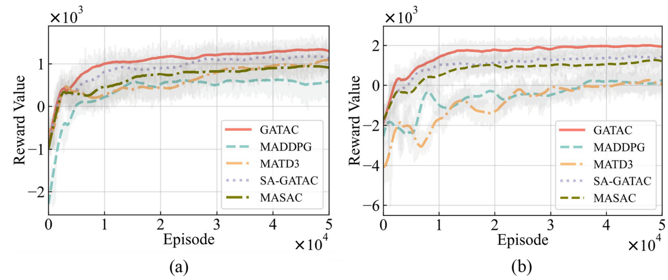


Fig. 7. Training reward curve comparisons. (a) One fixed-wing UAV and five multi-rotor UAVs. (b) One fixed-wing UAV and nine multi-rotor UAVs.

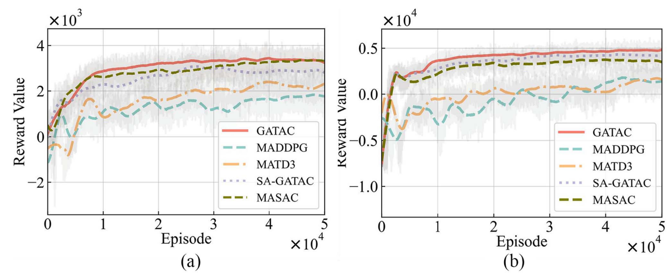


Fig. 8. Training reward curves comparisons. (a) Two fixed-wing UAVs and ten multi-rotor UAVs. (b) Two fixed-wing UAVs and 18 multi-rotor UAVs.

B. Simulation Results and Analysis

In this section, simulation results and analysis are presented. We introduce simulation scenarios and designs related experiments to compare the GATAC algorithm with above the four baseline algorithms.

1) Small Scale UAVs Swarm: In this experiment, Fig. 7(a) illustrates that the GATAC algorithm and baseline algorithms converge to the maximum reward value after approximately 30,000 rounds. Due to the relatively small scenario scale, the highest reward values of GATAC, MATD3, SA-GATAC and MASAC algorithms are similar. Fig. 7(b) shows that as the number of multi-rotor UAVs increases to nine, MATD3 and MADDPG achieve significantly lower rewards compared to GATAC, SA-GATAC and MASAC. This is because when increasing the total number of individual UAV swarm, the amount of interaction information between them increases. The GATAC algorithm can effectively identify and utilize the most relevant information, thereby improving the learning process and results, which is conducive to training optimal cooperative relationships.

Fig. 8 presents training reward curves of two UAV swarms for GATAC and four baseline algorithms. Two fixed-wing UAVs are set as central nodes, with scenes featuring 10 and 18 multi-rotor UAVs and 10 ground targets for search.

In Fig. 8, compared to Fig. 7, while the size of individual UAV swarms remains unchanged, the overall number of cooperative UAVs increases. The reward gap among the GATAC algorithm and both MADDPG and MATD3 further widens. This is due to the fact that the GATAC algorithm adopts clustering combinations, and the cooperation graph pre-sets the adjacency relationships between each fixed-wing UAV and multi-rotor UAV node. UAVs can learn more effective cooperation strategies under established rules.

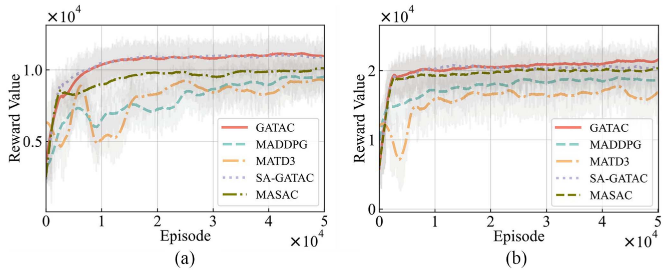


Fig. 9. Training reward curve comparisons of large scale UAVs swarms. (a) Four fixed-wing UAVs and 12 multi-rotor UAVs. (b) Eight fixed-wing UAVs and 24 multi-rotor UAVs.

2) Large-Scale UAV Swarms: Fig. 9 displays training reward curves for fixed-scale UAV swarms using GATAC and four baseline algorithms, with a fixed-wing to multi-rotor UAV ratio of 1:3. Four and six fixed-wing UAVs respectively serve as central nodes, alongside 12 and 24 multi-rotor UAVs and 20 ground targets for search scenarios respectively.

In Fig. 9(a), despite increasing the number of UAV swarms to four, GATAC, SA-GATAC and MASAC reward curves continue to converge steadily. However, the reward value of the MASAC algorithm is significantly lower than that of GATAC and SA-GATAC algorithms. Compared to the MASAC algorithm, after clustering with the GATAC algorithm, the size of individual UAV swarms is reduced and the combinations are fixed, resulting in a fixed amount of data input into the network, which improves the overall system adaptability. The GATAC algorithm assigns different weights to each neighboring node. This enables the model to adaptively focus on more important neighbors, thereby improving the effectiveness of information aggregation. whereas MATD3 and MADDPG algorithms show noticeable fluctuations, indicating compromised training effectiveness. Fig. 9(b) shows that the performance gap in convergence values between comparative algorithms diminishes as the number of UAV swarms increases to eight. This trend arises due to the limited simulation area size, where increased UAV numbers elevate the probability of random search. From these three sets of experimental figures, it is evident that while the training effectiveness of GATAC, SA-GATAC and MASAC algorithm is generally similar, the performance across tasks of different scales exists variance. Specifically, the graph attention model adopted by GATAC proves more suitable for larger-scale tasks.

3) Average Number of Found Targets: The simulation results are derived from 5,000 randomly generated test cases, with each algorithm evaluated under identical conditions. The ultimate goal of post disaster rescue is to search for more ground targets. Fig. 10 illustrates the comparison of target quantities explored by different algorithms in six experiments. The GATAC algorithm achieves a target search rate of over 68% in all scenarios. When a configuration of one fixed-wing UAV and five multi-rotor UAVs is used, target search rates of over 60% are not achieved by all other algorithms. When the total number of ground targets remains constant, the total number of targets searched increases with the number of UAVs. With a configuration of 24 multi-rotor UAVs and 8 fixed-wing UAVs, nearly all algorithms are able to search for over 80% of the targets, and the GATAC algorithm exceeded 85% search rate. The GATAC algorithm demonstrates strong performance by integrating entropy regularization, which enhances its exploration capabilities and allows for the search of a larger number of targets. Furthermore, the incorporation of attention mechanisms enables UAV swarms to maintain robust collaborative efficiency across varying scales.

4) Power Consumption: Fig. 11 presents the average energy consumption of UAVs across six experiments employing different algorithms. In tasks involving one fixed-wing and five multi-rotor UAVs, the MADDPG and MATD3 algorithms mistakenly focus on optimizing their own energy consumption to obtain higher rewards. The energy consumption performance of UAVs using the GATAC algorithm is not optimal when the number of UAVs is low, but surpasses others in tasks involving more than 12 UAVs. Although GATAC relies on regular entropy to increase exploration capability, it can still maintain low energy consumption because the algorithm has learned the appropriate level of exploration instead of blindly exploring. Compared with the MASAC algorithm, the GATAC algorithm can fly further and search for more ground targets with similar power consumption.

5) Broken Link Duration: In practical tasks, communication link interruptions in UAVs may result in permanent connectivity loss, potentially leading to substantial operational risks. Fig. 12 presents the average broken link duration of UAVs across six experiments employing different algorithms. In all scenarios, the GATAC algorithm has a safer and more stable average broken link duration between 1.0 and 3.8 seconds and is less affected by environmental changes. The MASAC, MADDPG, and MATD3 algorithms exhibit higher variability in average broken link duration in different scenarios. The longest broken link duration for each algorithm reaches 8.7, 12.9, and 11.7 seconds, respectively. And although the SA-GATAC algorithm performs more stably compared to these three algorithms, it is not as good as GATAC algorithm in general.

C. Experiment Design With Real-World Testing

In the previous sections, the performance of the GATAC was verified through software simulation. However, simulation environments struggle to fully replicate environmental disturbances and UAV dynamics. Therefore, to further validate algorithm practicality and transfer ability, this section utilizes a motion capture system to establish a multi-UAV validation platform. The proposed multi-UAV cooperative trajectory optimization algorithm is deployed and validated.

1) Platform Design Verification: The architecture of the multi-UAV cooperative verification platform is depicted in Fig. 13 and comprises the following key modules: motion capture positioning system, multi-UAV control system, cooperative trajectory optimization system, micro-UAVs, wireless UAV LAN, and wired switch.

Due to limited computational capabilities of the micro-UAV, deep reinforcement learning-based algorithms for real-time trajectory optimization cannot be directly deployed. Hence, this experimental platform adopts a centralized server controlling

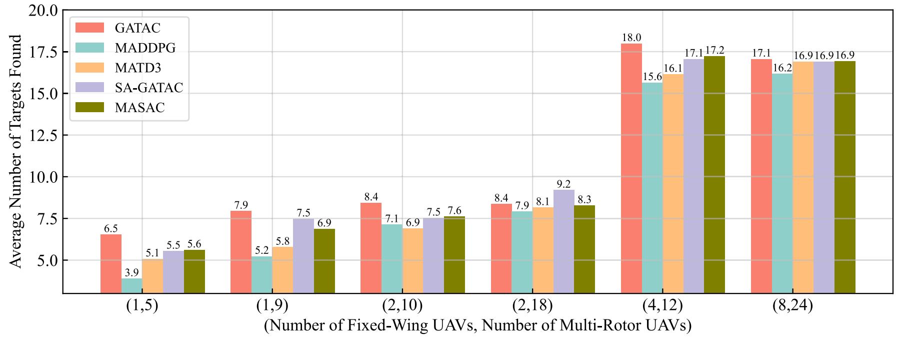


Fig. 10. Comparison chart of average number of targets found.

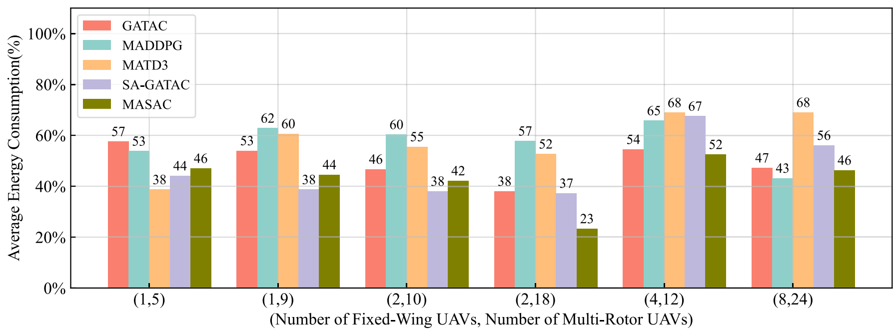


Fig. 11. Comparison of average energy consumption.

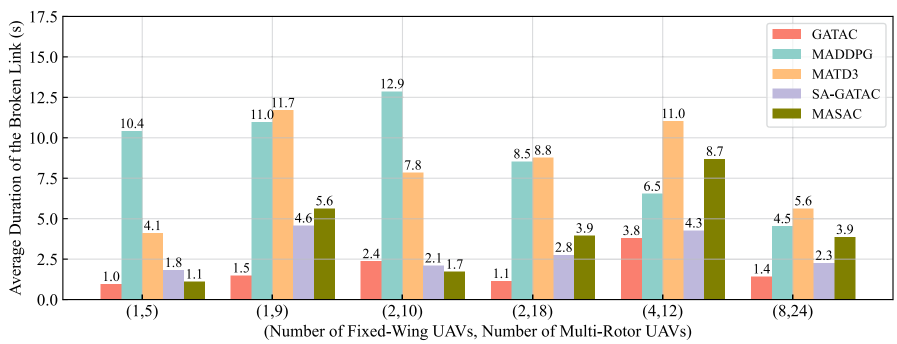


Fig. 12. Comparison of average broken link duration.

multiple UAV devices. Within the server, multiple independent subsystems simulate multi-agent systems, calculating and transmitting real-time desired UAV positions, velocities, and acceleration commands. While the operational platform is centralized, Actor networks can be distributed. Furthermore, during the execution phase, computational demands are modest, obviating the need for a GPU-equipped central control server. This facilitates future migration to UAV platforms with strong edge computing capabilities, enabling fully distributed collaborative deployment.

The motion capture positioning system employs a domestically produced solution from NOKOV Motion Capture System, which uses multiple cameras to simultaneously capture reflections from small spherical markers. Through upper-level software, the system calculates positions with sub-millimeter precision in three-dimensional space.

The cooperative trajectory optimization system consists primarily of previously proposed multi-UAV cooperative trajectory optimization algorithms, each deploying corresponding algorithm models in the server according to the experimental scenario. Initially, the cooperative trajectory optimization system acquires multi-UAV status information from the motion capture system using ROS message packets. Network communication status, ground user, and target search information are simulated by software and mapped to real-world scenarios. Subsequently, these status information inputs to the Actor network produce corresponding UAV actions. Finally, by employing UAV dynamic models, the system calculates task strategies such as expected UAV positions and velocities for each timeslot.

2) Experimental Design: The experimental site is shown in Fig. 14(a). This experiment simulates realistic scenarios of indoor UAV operations, necessitating proportional reduction of scene scale. Flight duration is set to 30 seconds, with a safe flight distance of 0.2 meters, and UAVs are required to effectively avoid obstacles. The miniature UAVs used in experiments lack actual network coverage and target detection capabilities.

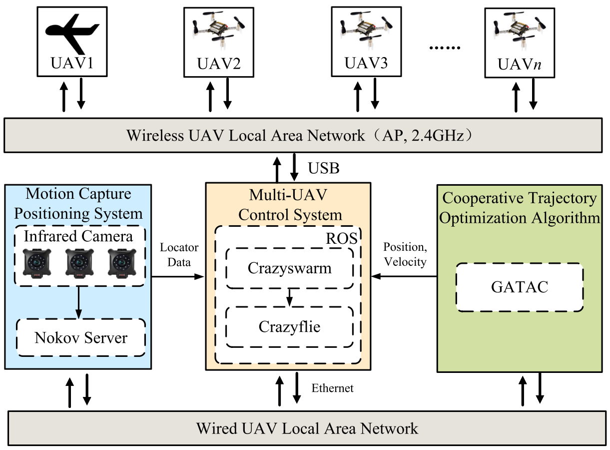
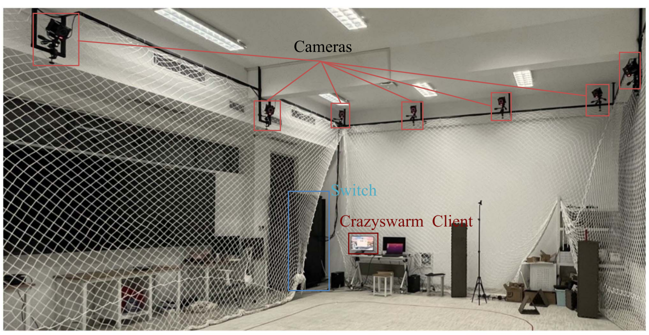
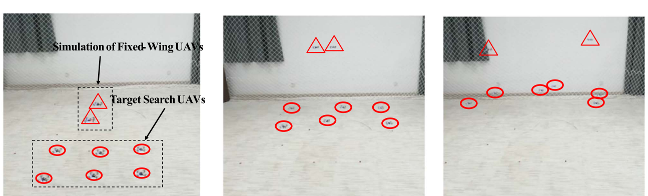


Fig. 13. Experiment platform architecture of multi-UAV cooperation.



(a)



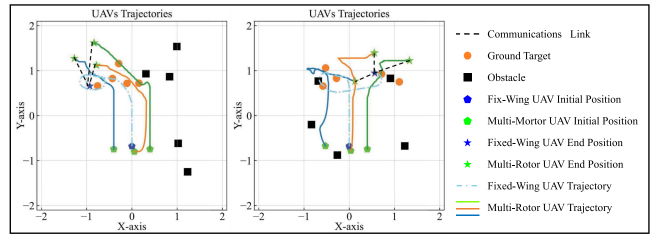
(b) (c) (d)

Fig. 14. Indoor physical simulation. (a) Indoor motion capture and positioning system. (b) UAVs starting positions. (c) UAVs searching positions. (d) UAVs completing positions.

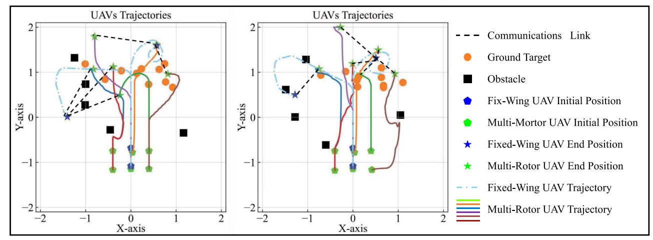
Therefore, the ground users, search targets, and obstacles required for the experiments are all mapped to the real scenario through simulation methods. Additionally, the networking model and dynamic model of the UAVs have also been simulated.

The starting positions for the multi-UAV target search task are depicted in Fig. 14(b), with triangular markers indicating simulated fixed-wing UAVs flying at slightly higher altitudes, continuously circling to provide network coverage for low-altitude search UAVs. The Fig. 14(c) shows the UAVs are searching for targets. After completing the target search task, the final positions of the UAVs are shown in Fig. 14(d).

This experiment adopts a leader-follower cooperation model. Due to limited indoor space, fixed-wing UAVs cannot be deployed, thus multi-rotor UAVs simulate fixed-wing UAVs by incorporating fixed-wing dynamic constraints and restricting



(a)



(b)

Fig. 15. The target search trajectory of fixed-wing and multi-rotor UAVs. (a) One fixed-wing and three multi-rotor UAVs. (b) Two fixed-wing and six multi-rotor UAVs.

minimum cruise speeds. Simulated fixed-wing UAVs circle at higher altitudes, while multi-rotor UAVs search targets at low altitudes. Two sets of experiments are conducted: one set with one fixed-wing UAV and 3 multi-rotor UAVs searching for 5 ground targets; another set with 2 fixed-wing UAVs and 6 multi-rotor UAVs searching for 10 ground targets, each set featuring 10 ground obstacles.

D. Analysis of Real-World Experimental Results

In the multi-UAV real-world validation experiments, the Crazyswarm system first establishes communication with each UAV, performs system self-checks, and matches with the motion capture positioning system. Subsequently, it commands the UAVs to take off and reach designated cruising altitudes. Once UAVs reach the starting points, the cooperative trajectory optimization system takes over, sending commands for desired UAV positions, velocities, and attitudes at each timeslot. Upon task completion, the Crazyswarm system directs UAVs to return to starting positions, issues landing commands, and locks UAVs after landing.

Fig. 15 illustrates the the heterogeneous multi-UAV target search experiment trajectories. The fixed-wing UAV, acting as the leader, is capable of swiftly reaching and hovering over the airspace with the highest number of target points. Meanwhile, the multi-rotor UAVs, functioning as followers, can rapidly search through all target points while simultaneously avoiding ground obstacles. The multi-rotor UAVs efficiently search all target points while avoiding ground obstacles. Additionally, the larger turning radius of fixed-wing UAVs is crucial for their efficient cruising trajectories during search tasks. As shown in Fig. 15(a), the search trajectories of a fixed-wing UAV and three multi-rotor UAVs are depicted in two different scenarios, fixed-wing UAVs first fly near the center of target distribution, followed by continuous circling, while multi-rotor UAVs maintain communication coverage and search ground targets within the coverage area. Fig. 15(b) illustrates the search trajectories of two fixed-wing UAVs and six multi-rotor UAVs, showing that larger swarm sizes are also capable of completing target search tasks. One UAV swarm tends to hover and search smaller and nearby targets, while the other swarm conducts broader searches for distant targets.

VII. Conclusion

Multi-UAVs have been widely employed in post-disaster emergency rescue missions due to their agility and ease of deployment. In this paper a leader-follower cooperative strategy has been implemented, with fixed-wing UAVs leading and multi-rotor UAVs following to carry out rescue missions. Next, a cooperative graph has been developed to model the relationships among neighboring UAVs, and GAT have been utilized to aggregate information from adjacent nodes. This approach reduces redundant information exchange among UAVs, thereby enhancing the scalability of the cooperative system. Then indices for cooperation importance and target importance have been designed to streamline the observation of state vectors. Afterwards, we have proposed the GATAC algorithm based on GAT and MARL. Simulation results have shown that the GATAC algorithm outperforms existing methods in terms of target identification and energy efficiency. Finally, a multi-UAV cooperative validation platform has been established, and experimental trajectory plots have the stable performance of the GATAC algorithm in real-world conditions.

In addition, the GATAC algorithm will also be applied to environmental monitoring and wilderness search and rescue missions in the future. In the leader-follower model, multiple multi-rotor UAVs follow a fixed-wing UAV to perform search tasks. This heterogeneous UAV swarm combines the advantages of both types of UAVs. However, if the leader fails, it may lead to the failure of the entire UAV fleet task centered around them. To address this concern, multiple leaders can be used during task execution. After a leader of the multi-rotor UAVs fails, the fixed-wing UAV in the fleet can be assigned to a new leader.

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